

mei

Metabolic Intelligence: Budget-Native Compute from Tag to Campus

David Charlot, Open Interface Engineering

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Measurement. Research study, not a journal final. Energy figures from the software reference are OpCounter analytical estimates, not board power. We have not synthesized or metered an FPGA board. No fabricated citations.

Designed for Klere (klere.ai). Energy is the binding constraint. The metabolic envelope is the physical condition under which the superior answer is obtained — not a cheaper answer, and not a Faustian trade. Edge and a resource-optimized central datacenter, not an uncapped hyperscale campus. Dedicated water, power, and cooling plants are not required by design (pack evidence: microwatt-class tags; grid-following lean campus; not invented liters). Research owns Metabolic Intelligence. Klere owns the product. Shannon, Landauer, and Value of Information are settled law. Estimates are not board measured_j.

Abstract

Metabolic Intelligence is a budget-native class of computer intelligence. The envelope is the physical condition under which the superior answer is obtained. It is not a Faustian choice, and it is not the cheapest answer. Answers that were thought impossible become reachable by metabolic information science: budget-native obtain, digital enzymes, physics-informed features, and test-time close. Grid levers and routing do not degrade the certified answer. They refuse or defer work that would violate the envelope. The answer that commits is the best one. The class is not a transformer update. It sits with Logical Intelligence (energy-based settle), World Labs (spatial world models), and other post-transformer programs as a next wave: intelligence whose first coordinate is the joule envelope, not the next-token corridor.

Designed for Klere. Metabolic Intelligence is designed for **Klere** (klere.ai) — Open Interface Engineering's product home for the MEI class. This research study owns the law and class definition. Klere owns the product embodiment. As of this writing the public site is a thin access / energy-efficient AI framing; it is not a shipped feature catalog. Tag, campus, and obtain-stack surfaces named below are **Klere roadmap** unless this study labels them as a **proven pack**. Product thesis in

one line: sell **edge** and **resource-optimized central** under an explicit joule envelope — not an uncapped transformer hyperscale campus — and keep proven-pack evidence separate from roadmap packaging. No invented meters, board measured_j, or liters of water saved.

Deployment class. MEI is designed for the **edge** and for the **resource-optimized central datacenter** — a lean, grid-following, budget-capped plant — **not** for the uncapped hyperscale campus that presupposes dedicated cooling loops, dedicated generation, and large ecosystem water/power uptake. At the energy budgets this class targets, those hyperscale support plants are **not required**: that is a **design consequence** of budget-native operation, grounded in pack evidence (μ W-class tags under a metabolic schedule; grid-following lean campus with ordered levers). It is **not** a claim of measured liters of water saved, nor a board measured_j reading. Estimates remain estimates; soft-ref path: board_synth_claimed=false.

Class properties, stated plainly:

1. **Physics-informed.** Features and actuators respect sensor kinetics, heater physics, Landauer floors, and grid constraints as settled law, not optional add-ons.
2. **Test-time inference.** Binding, routing, and lever selection happen under the live budget; the decision is shaped at serve time.
3. **Super learning / sequential learning.** New contexts and classes can be absorbed under the same budget without a full pretrain replay.
4. **Zero-shot capable under grammar.** When Lookup or Formula covers the coordinate, the system answers without opening a generative leaf.
5. **Not pre-training required** as the default path. Abundance updates, family indices, and obtain-routers learn and decide under local cost; large corpus pretrain is residual, not the substrate.

Dual-scale proof in this study (edge + resource-optimized central — not uncapped hyperscale):

- **Edge / tag scale.** Digital enzymes on cold-chain gas tags. Cortex-M4 instruction counts (emulator-measured): enzyme families with early exit take 1,723 instructions vs 2,451 for an int8 MLP at matched accuracy. On SmellNet sequential learning (50 foods, five groups, no revisit), enzymes keep 77.7% vs 18.4% for the MLP. A metabolic heating schedule draws about 10 μ W total (datasheet + schedule model): heater about 3.0 μ W, MCU sleep about 6.9 μ W on STM32U3; about eight years on a CR2032. Classifier decisions are about 92 nJ; the sleep floor dominates on the metabolic schedule. At this envelope, dedicated cooling plants and campus-scale water loops are not in the design — coin-cell physics is the plant. Chicken-spoilage Anwar results are internal only and are **not published** here (no licence).
- **Resource-optimized central (lean campus).** A metabolic grid service for a **budget-capped** 100 MW AI plant — grid-following, not uncapped hyperscale — tested end to end against a real OpenADR 3.1 VTN and a real MQTT DSX Flex broker. Plant is a model; protocols are real. With node sleep available: zero time over the grid cap across the closed scenario set. Headline lever order: pause deferral → slow per-user speed → route smaller model → lower effort → battery → shed. The service is how a lean central stays inside shared power and water ecosystems without requiring a private uncapped cooling/power campus.

Metabolic Intelligence (MEI once in this abstract) owns the **budget envelope and actuators** that make companion floors bind. **Mixture of Limits** chooses the gear. **Notational Intelligence** owns certify-before-commit. **Satiation** owns stop when Value of Information (Vol) is zero or budget refuse fires. This study owns sleep/heater schedules, OpenADR/MQTT service levers, and the (J(mlq)) obtain-router that prices look up → calculate → model → guess under λ at stake.

Hardware dual-map: the class runs today on GPUs and MCUs (von Neumann), and points at post-von Neumann substrates (neuromorphic, analog / in-memory, photonic, reversible / adiabatic) where energy-based and latent settle map natively. Estimates remain estimates. Soft-ref path:
board_synth_claimed=false; package measured_j only when Metered.

Why this is the future

Industry and academic messaging still treats transformer scale, plus the hyperscale power and water plant that scale presupposes, as the default future of computer intelligence. That default can be answered. It cannot be answered by a slogan. A skeptic who wants to set Metabolic Intelligence aside has to deny settled physics (Shannon, Landauer, or Value of Information), deny a labeled measurement from this pack, or deny an explicit roadmap this study already separates from proven pack. This section states the argument. Appendix F answers the dismissals that usually follow. Living diagram: </living/mei/#mei-arg-01>.

Energy is the binding constraint. Shannon (1948) gives bits a rate-distortion structure: past a fidelity point, more bits buy little additional fidelity. Howard's Value of Information (1966) prices an observation by the improvement it makes in decision utility, net of cost. When that value is zero on a stated completeness predicate, further spend does not raise the written objective. Landauer (1961) prices irreversible bit erasure at $(k_B T)$ per bit in the ideal model. On a receipt that quantity is a labeled estimate, not a wattmeter. Horowitz (2014) shows that practical CMOS energy is dominated by moving data, far above that ideal. These are settled coordinates. They are not Open Interface Engineering inventions. A future whose first coordinate is uncapped watts has to say which of them fails.

The transformer corridor plateaus as a deployment future. The corridor has produced real products: pretrain a large generative model, then spend tokens and watts at serve time. Quality inside that corridor can keep moving. The plateau is not a claim that models stopped improving, and it is not a fabricated scaling chart. The plateau is this: once a chore is complete, extra tokens do not raise the written objective (Satiation). Past the rate-distortion point, extra bits buy little decision utility (Shannon). The plant underneath still presupposes dedicated cooling, dedicated generation, and large ecosystem water and power. Uncapped import stops being the coordinate that buys the next useful answer. Logical Intelligence, V-JEPA, and World Labs are real post-transformer programs on other axes (constraint commit, latent prediction, spatial world models). They do not make that import optional, and this study does not speak for their results.

The envelope obtains the superior answer. Let (B) be the metabolic envelope: joules on a tag episode, or watts on a lean-campus import cap. (B) is the physical condition of the answer, not a discount on it. Metabolic information science — budget-native obtain, enzymes, physics-informed features, test-time close — is how answers that were thought impossible become reachable. Mixture of Limits chooses the gear: Lookup, then Formula, then Solver, then Model last. The obtain-router prices the gear as $(J(mlq))$ before spend: energy plus a stake-weighted uncertainty term. The estimator is Lookup or Formula and is capped so meta-compute cannot eat (B). If that estimate exceeds the cap, the router takes the default gear and does not run a heavy estimator. Binding and lever choice happen at test time inside the live envelope. Levers and routing refuse or defer work that would violate the envelope. They do not degrade the certified answer. The answer that commits is the best one. On the digital-enzyme path, new classes can be absorbed without a full pretrain replay. That sentence is path-scoped (Proposition 7). It is not a claim about V-JEPA or World Labs, which pretrain on large video, and it is not a claim that Logical Intelligence skips pretrain.

Dual deployment removes the hyperscale water and power prerequisite. The class is designed for two plants, not for an uncapped hyperscale campus. At the edge, the plant is a coin cell: a metabolic heating schedule on a cold-chain gas tag, datasheet plus schedule model, about $10\ \mu\text{W}$ class, with MCU sleep dominating heater average power on that schedule (Proposition 3). Instruction counts are emulator-measured (1,723 for enzyme families with early exit versus 2,451 for an int8 MLP at matched accuracy). Sequential learning on SmellNet is a real dataset protocol (enzymes 77.7% versus an MLP at 18.4% across 50 foods, five groups, no revisit). At the resource-optimized central, the plant is a budget-capped, grid-following lean campus: real OpenADR 3.1 and a real MQTT DSX Flex broker, with a plant model for the watts. With node sleep available, the closed scenario set records zero time over the grid cap (Proposition 5). Because those envelopes are the design, dedicated water plants, dedicated power plants, and hyperscale cooling loops are not required. That is a design consequence of the class plus pack evidence. It is not a measured liter count. It is not a board measured_j.

Klere is the product home. This study owns the law and the class definition. Klere (klere.ai) owns product embodiment. As of this writing the public site is thin access framing, not a shipped feature catalog. Tag, lean-campus, and obtain-stack surfaces are Klere roadmap unless this study labels them proven pack (Appendix E). Computers are hardware. Software is applied engineering under the envelope. Method is the claim axis. Geography is not.

A reader who grants Shannon, Landauer, and Value of Information, and who grants the pack labels above, still has to name a future in which uncapped hyperscale import is required for the next useful answer. Appendix F is the set of ways that denial is usually phrased, and the evidence class of each reply. Soft-ref path: board_synth_claimed=false.

1. Introduction: budget-native intelligence

People should not have to fight over energy to eat or to use AI. That civic sentence is the design brief. Cold-chain tags that watch food should last years on a coin cell. Central plants that serve answers should follow the grid when the grid asks them to cut, without inventing false board joules or collapsing every research law into one paper.

Uncapped hyperscale campuses pull dedicated cooling, dedicated generation, and large ecosystem water and power. Metabolic Intelligence is **not** that deployment class. It is designed for the **edge** and for the **resource-optimized central datacenter**: a lean, budget-capped, grid-following plant. Because the class runs at lower energy budgets by construction, dedicated water plants, dedicated power plants, and hyperscale cooling loops are **not required** — stated as a **design consequence** of the budget-native envelope, with pack evidence at both ends (μW tags under a metabolic schedule; OpenADR/DSX lean-campus levers). Do **not** invent liters of water saved or board measured_j from that consequence. Soft-ref path: board_synth_claimed=false.

The dominant construction for computer intelligence still treats the transformer corridor as the substrate: pretrain a large generative model, then spend tokens and watts at serve time. That corridor has produced real products. It is not the only class. **Logical Intelligence** casts reasoning as energy minimization over constraints (Business Wire, 20 Jan 2026). **World Labs** casts spatial intelligence as world models that predict views and dynamics (Atlas, 1 Sep 2026). **Metabolic Intelligence** casts intelligence as **budget-native obtain and actuate**: the envelope is the physical condition under which the superior answer is obtained. It is not the cheapest answer. The class is designed for **Klere** (klere.ai): research owns the law; Klere owns the product embodiment. Do not read the thin public landing page as live product features.

Shannon (1948) prices bits under uncertainty as settled law. Howard's Value of Information (1966) prices whether another observation is worth its cost for a decision. Landauer (1961) prices irreversible bit erasure in joules at temperature (T): ($E_{\text{bit}} = k_B T$) per bit erased in the ideal model. On receipts that Landauer quantity is a **labeled estimate**, not a wattmeter reading. Together they imply floors: past a point, more bits stop buying outcomes that matter for a stated benefit. Metabolic Intelligence embodies those floors as **schedules and actuators** at the edge of a tag and at the meter of a lean central plant.

Formal thesis. Let (B) be the metabolic envelope in joules (tag episode) or watts / megawatts (lean-campus import). (B) is the physical condition under which the superior answer is obtained. It is not a menu of cheaper substitutes. Metabolic Intelligence returns that answer by budget-native obtain, enzymes, physics-informed features, and test-time close. A stated utility (U) (accuracy, completeness of a chore, interactive close) is what the committed answer maximizes. Grid-following levers refuse or defer work that would violate (B). They do not degrade the certified answer. The answer that commits is the best one. The substrate is digital enzymes and other algorithmic compute under that envelope, plus a grid-following service whose levers keep the envelope. Deployment scopes are **edge** and **resource-optimized central** — not an uncapped hyperscale campus whose first coordinate is unlimited import. Class properties follow: physics-informed features and heaters; test-time binding

and routing; sequential / super learning without full pretrain replay; zero-shot close when grammar covers; pretrain as residual leaf, not default. Computers are hardware; software is applied engineering under constraints.

Three measurement facts constrain every later number. First, curve and ledger watt-hours are **model / measured-kernel / looked-up** composites, labeled by evidence class. Second, tag microwatts mix **datasheet** and **schedule model**; instruction counts are **emulator-measured**. Third, `board_synth_claimed=false`; package `measured_j` appears only when a labeled meter returns a reading.

2. Formal problem, definitions, and propositions

2.1 Problem statement

Problem (Budget-native obtain). Given a need (q) that feeds a decision (y), an explicit budget (B), and a menu of obtain mechanisms ($M = \{, , , \}$) with costs and uncertainty, select mechanism ($m^\wedge$) and actuation schedule ($s^\wedge$) so the committed answer is the superior one under the envelope (B). Work that would violate (B) is refused or deferred. Stop when further spend does not raise a stated completeness predicate (Satiation) or when policy refuse fires (Notational Intelligence). The envelope is not permission to return a worse answer.

The problem is dual-scale by construction — **edge** and **resource-optimized central**, not uncapped hyperscale:

Scale	Budget unit	Actuators	Success metric in this study
Edge / tag	μ W continuous; nJ–mJ per event	Heater duty, MCU sleep, enzyme binding / early exit	Lifetime on CR2032; decisions under schedule; sequential accuracy
Resource-optimized central (lean campus)	MW import cap (grid-following; not uncapped)	Pause, speed, route, effort, battery, shed	Time over cap; interactive served share

2.2 Definitions

Term	Definition in this paper
Metabolic Intelligence (MEI)	Budget-native class. The metabolic envelope is the physical condition under which the superior answer is obtained, at the edge and the resource-optimized central (not uncapped hyperscale). Not a cheaper answer.
Budget envelope	Explicit joule or power condition (B) under which the superior answer is obtained. First-class physical

Term	Definition in this paper
	condition, not a post-hoc telemetry label and not a quality discount.
Digital enzyme	Silicon recognition unit with binding site, product, allosteric context switch, Hill cooperativity, abundance as learned weight, and a cost for every binding check. Bit-enzyme variant uses no multiplies.
Metabolic schedule	Duty cycle for sensing and heating that meets a monitoring objective while minimizing average power (heater + sleep + radio).
Obtain mechanism	How an input is acquired: Lookup, Formula / Calculate, Solver, Model, Guess — ordered by Mixture of Limits; priced by $J(\text{mlq})$.
Obtain-router	The rule ($m^* = _m J(\text{mlq})$) that selects the mechanism under λ at stake.
Grid-following service	Control plane that accepts OpenADR / DSX Flex signals and applies ordered levers so campus import stays under cap.
Evidence class	Label on a number: real protocol, emulator-measured, datasheet, measured-kernel model, looked-up, calculation, plant model, estimate (M/C/V/R/D/S/P in the pack radar).
Soft-ref path	<code>board_synth_claimed=false</code> ; estimates $\neq$ package measured_j.

2.3 Settled law: Shannon, Landauer, Vol

These are not OpenIE inventions. They are settled coordinates this study *uses*.

Law	Statement used here	Role in MEI
Shannon (1948)	Bits under uncertainty have a rate–distortion structure; more bits past a point buy little for a fixed fidelity.	Justifies Lookup / Formula before Model when the coordinate is already known.
Landauer (1961)	Irreversible bit erasure has an ideal floor ($k_B T$) per bit.	Labeled estimate on receipts; not a wattmeter. Motivates treating joules as scarce even when digital inference is free at the money margin.
Howard Vol (1966)	Value of Information is the expected improvement in decision utility from an observation, net of cost.	When $\text{Vol} \leq 0$ under λ , Satiation / Mixture of Limits name stop; MEI makes stop executable as refuse of heater pulse or campus lever.

Horowitz (2014) documents that practical CMOS energy is dominated by data movement and memory, far above the Landauer ideal. That gap is why Metabolic Intelligence schedules heaters and campus import meters rather than claiming thermodynamic optimality from whiteboard erasure.

2.4 Propositions (claims with evidence class)

Each proposition is a claim. Evidence class is stated. Peer overlap is acknowledged; uniqueness is scoped.

Proposition 1 (Envelope primacy). *An intelligence class whose first coordinate is an explicit budget envelope can bind obtain and actuation at microwatt (edge) and megawatt (resource-optimized central) scales under one thesis — without presupposing uncapped hyperscale cooling/power plants.* Evidence: this study's dual-scale pack (tag schedule + OpenADR lean-campus service). Class: **real protocol** (campus) + **datasheet / schedule model / emulator** (tag). Peer slices: Dynamo power caps (campus only); TinyML (edge only). Unique stack claim: dual scale + OpenADR/DSX + (J(mlq)) + enzymes + Satiation stop. Ecosystem-resource avoidance (water / dedicated power / hyperscale cooling) is stated as a **design consequence** of that envelope, not as measured liters saved.

Proposition 2 (Enzyme sequential learning). *Under a fixed MCU budget, digital enzymes absorb sequential class groups with far less catastrophic forgetting than an int8 MLP of matched decision cost.* Evidence: SmellNet 50 foods, five groups, no revisit — enzymes 77.7% vs MLP 18.4% (**real dataset protocol**). Peer overlap: AdaTM continual Tsetlin; classical Hopfield / DenseAM pattern add. Do **not** claim universal end of catastrophic forgetting.

Proposition 3 (Sleep floor on metabolic schedule). *On a cold-chain gas tag with metabolic heating (~9.6 readings/day, Oct 2026 parts), MCU sleep current dominates heater average power.* Evidence: BME690 heater ~3.0 μ W vs STM32U3 sleep ~6.9 μ W (**datasheet + schedule model**). Correction: "heater is always the cost" is false on this schedule.

Proposition 4 ((J(mlq)) obtain law). *Pricing obtain mechanisms by energy plus λ -weighted uncertainty selects Lookup/Formula when they measure the same quantity and reserves Model for residual leaves.* Evidence: synthesis ledger math + sourced inputs (**calculation + looked-up**); not board measured_j. Peer: PEARL / OmniRouter price *which* LLM; they do not price *which* obtain zone.

Proposition 5 (Campus cap with node sleep). *A 100 MW campus plant model under real OpenADR 3.1 and real DSX Flex MQTT, with node sleep available, records zero time over the grid cap across the closed scenario set.* Evidence: **real protocol + plant model**. Without node sleep, idle GPUs alone can exceed a deep cut (scenario E: 96 min over cap).

Proposition 6 (Free-at-margin $\neq$ free joules). *Published collapse in money price of digital inference does not identify free energy, free actuation, or unbounded value after a chore is complete.* Evidence: companion [Satiation](#) (Epoch series cited there); Landauer / Horowitz as floors. MEI supplies the envelope actuators that make the economic stop physical.

Proposition 7 (Class properties are path-scoped). *No-pretrain, zero-shot, and test-time claims must be scoped by path (enzyme vs latent vs certify EBM). Mixing scopes invents a contradiction.* Evidence: V-JEPA 2-AC zero-shot planning requires large video pretrain (Assran et al., 2025); enzyme online path does not. Logical Intelligence constraint energy $\neq$ campus joules (Business Wire, 2026).

2.5 Formal lemmas (stated, not board-proved)

Lemma A (Obtain dominance under matched measurement). If two mechanisms (m_1, m_2) measure the same quantity (q) and ($E_{\{m_1\}} < E_{\{m_2\}}$) with ($\{m_1\}\{m_2\}$), then ($J(m_1|q) < J(m_2|q)$) for all (λ, S_q). *Proof sketch:* both additive terms weakly favor (m_1). Evidence class: **calculation** (definition of (J)). Implication: Lookup beats Calculate when the table holds the same coordinate; Calculate beats Model when a closed form exists.

Lemma B (Precision sizing). For fixed (λ, S_q) and target excess cost (λ), the uncertainty that equalizes the λ -term with (λ) is ($\lambda^* = \lambda$) (ignoring staleness). Spending below (λ^*) wastes energy; spending above leaves decision risk on the table. Evidence: **calculation**. Implication: satiation of precision is a first-class stop, not a soft habit.

Lemma C (Cache refresh). Under linear staleness ($\lambda_q t$), the refresh interval that balances obtain energy against λ -weighted variance growth is ($t^* = 2E / (S_q \lambda^2)$) when (p_m) is small. Evidence: **calculation**. Implication: metabolic schedules on tags and cache TTLs on campuses are the same object at different λ .

Lemma D (Cap feasibility without sleep). If idle import ($P_{\{ \}}$) exceeds the grid cap ($P_{\{ \}}$), no combination of speed / route / effort levers that leaves nodes powered can meet the cap. Scenario E is the concrete instance (96 min over cap without node sleep). Evidence: **plant model + real protocol**. Implication: node sleep (or equivalent power-down) is not an optional luxury at deep cuts.

These lemmas are engineering consequences of the definitions. They are not theorems about AGI. Soft-ref path: `board_synth_claimed=false`.

2.6 Related work by method (academic corridor)

Method is the claim axis. Geography is not. What this class absorbs is stated as law. Everything else in this section is cited only. URLs and dates below were checked on 2 Oct 2026. A vendor joule, a vendor latency, or a vendor accuracy stays on the vendor's page. It is not this pack's `measured_j`.

Absorbed (not citations). Four objects are Metabolic Intelligence.

1. **Physics envelope.** (B) is joules on a tag episode or watts on a lean-campus import cap. Energy budget is physics: the condition under which the answer is obtained. It is not a discount on the answer, and it is not a Faustian cheaper-answer trade.

2. **Superior answer.** Levers and routing refuse or defer work that would violate (B). They do not degrade the certified answer. The answer that commits is the best one. Budget-native obtain, digital enzymes, physics-informed features, and test-time close are how answers that were thought impossible become reachable. That reach is inside the envelope.
3. **Digital enzymes.** Binding site, product, allosteric context, abundance as learned weight, a cost on every binding check, early exit. The unit is this study (§5): emulator-measured Cortex-M4 counts and the SmellNet sequential protocol. Tsetlin machines and hyperdimensional computing, cited below, are edge cousins. They are not the enzyme.
4. **Dual hardware map.** The class runs today on GPUs and MCUs. It points at neuromorphic, analog in-memory, photonic, and adiabatic substrates where energy-based and latent settle can map (Appendix B). The map is operational. A substrate paper is not a second definition of the class. Deployment is the edge and the resource-optimized central. Hyperscale water, power, and cooling plants are not required by design.

Cited only.

Energy-based models, Hopfield networks, and Dense Associative Memory. Hopfield (1982) casts associative memory as descent on an energy. Krotov and Hopfield (arXiv:1606.01164, 2016) raise capacity with higher-order energies (Dense Associative Memory). Ramsauer et al. (arXiv:2008.02217, 2020) recover softmax attention as a modern Hopfield update. Hoover, Krotov, and coauthors (arXiv:2302.07253, NeurIPS 2023) define the Energy Transformer: attention layers built to minimize a token energy. Gladstone et al. (arXiv:2507.02092, 2025) train Energy-Based Transformers that assign an energy to an input–prediction pair and predict by gradient minimization of that energy. Bacvanski, Hopfield, Krotov, and coauthors (arXiv:2512.15002, 16 Dec 2025) propose analog RC and crossbar circuits for Dense Associative Memory and analyze constant-time inference from amplifier specifications; that bound is their analysis, not a meter here. Montanari, Bullo, Krotov, and Motter (arXiv:2604.05042, 2026) survey energy-based dynamical models from Hopfield networks through Dense Associative Memory to oscillator machines. Logical Intelligence is the product neighbor on this axis, not these theory papers. Bodnia and Hanin (21 Jan 2026) describe Kona as non-autoregressive at the trace, globally scored, and continuous-state: an energy over partial and complete traces, with Aleph coordinating calls to Kona and language models. Kona 1.0 pilots are the Business Wire account of 20 Jan 2026. Their scalar is constraint energy. It is not campus import joules and not a coin-cell schedule. This study does not adopt their benchmark figures.

Latent prediction and spatial world models. LeCun (version 0.9.2, 27 Jun 2022) names the joint-embedding predictive architecture as a path to autonomous machine intelligence. V-JEPA 2 and V-JEPA 2-AC (Assran et al., arXiv:2506.09985, 11 Jun 2025) predict in a latent video space and, after large video pretraining plus less than 62 hours of unlabeled robot video, plan a robot arm from image goals without a task reward. World Labs Marble (12 Nov 2025) is a generative multimodal world model. Atlas (1 Sep 2026) is their later omni model over text, images, video, and 3D. A latent model is an optional Model-last leaf under Mixture of Limits. Their pretrain stays on their path (Proposition 7). This study does not speak for their results.

State-space efficiency corridor. Mamba (Gu and Dao, arXiv:2312.00752, 2023) is a selective state-space sequence model: linear scaling in length, constant-size recurrent state, no attention block. Mamba-2 (Dao and Gu, arXiv:2405.21060, 2024) states a structured state-space duality with attention. The corridor prices sequence compute inside the transformer-replacement race. It does not price an obtain zone, a heater duty cycle, or a grid cap.

Neuromorphic, analog, and photonic substrates. Intel announced Hala Point on 17 Apr 2024: 1,152 Loihi 2 processors, 1.15 billion neurons, stated maximum 2,600 W. That watt figure is Intel's. Chen et al. (Nature Communications, 30 Sep 2025, doi:10.1038/s41467-025-63794-4) demonstrated an ALBERT transformer on a 14 nm analog in-memory inference chip. Ahmed et al. (Nature, 9 Apr 2025, doi:10.1038/s41586-025-08854-x) report a photonic processor executing ResNet, BERT, and an Atari reinforcement-learning workload. These demonstrations sit on the dual hardware map. They are not the class. Shipping neuromorphic gas or odour nanojoule figures were not found. Innatera Pulsar remains a vendor milliwatt-class audio and radar part, above the tag schedule near 10 μ W (datasheet plus schedule model).

Tsetlin machines and hyperdimensional computing. Granmo (arXiv:1804.01508, 2018) learns propositional clauses with teams of Tsetlin automata. Kanerva (*Cognitive Computation*, 2009) defines hyperdimensional computing on high-dimensional random vectors. Tunheim et al. (arXiv:2501.19347, 31 Jan 2025) report an all-digital 65 nm Tsetlin accelerator at 8.6 nJ per MNIST frame. Hyperdimensional Tsetlin machines (arXiv:2406.02648, 2024) combine the two encodings. Cite them as sparse edge cousins: online learning and nJ-class decisions on the tasks they publish. They do not supply the OpenADR obtain-router or the dual-scale thesis.

Open post-training recipes. Trillium Labs (trilliumlabs.org), Nathan Lambert and Tom Zick, was unveiled on 2 Oct 2026. Lambert's note the same day says the nonprofit is building open post-training recipes and intends to expand into open infrastructure for recursive self-improvement, reward hacking, and multi-agent systems. WIRED the same day (Will Knight) reports the initial focus as post-training: publishing experiment details so others can study and replicate them. Those sources describe a program. They do not contain a recipe, a dataset, a configuration, or a model release. None is inferred. Trillium Labs is the open corridor of the frontier transformer stack. It is not Metabolic Intelligence and not a Klere product.

Envelope actuators, not a class. PEARL, OmniRouter, and Energy-Aware LRM (2025–Jan 2026) price which language model answers. NVIDIA Dynamo 1.5 (18 Sep 2026), InferenceX, MLPerf, Emerald DSX Flex, and the AI Energy Management Alliance are serving and grid actuators (§11.4). Metabolic Intelligence binds those actuator shapes to Mixture of Limits obtain zones. The actuators are not the intelligence class.

Companion law stays companion law, named in full: Mixture of Limits navigates, Notational Intelligence commits, Satiation stops. This study is the envelope those floors spend.

2.7 Evidence grammar (how to read every table)

Tag in this paper	Meaning	When it can support a sell claim
real protocol	Talked to a real VTN / broker / schema	Protocol interoperability yes; plant watts no
emulator-measured	Cortex-M4 instruction counts from ELF under emulator	Relative enzyme vs MLP cost yes; board joules no
datasheet	Vendor electrical tables	Schedule model inputs
measured-kernel model	AI Configurator kernels + published power ratios	Wh/answer bands as estimates
looked-up	Artificial Analysis medians, public blogs	Baseline speeds
calculation	Central differences, (J) algebra, CoreMark→nJ	Algebraic consequences
plant model	Campus thermal/power simulator	Scenario comparisons under stated assumptions
estimate	Ledger fuse, capacity life	Always banded; never measured_j

board_synth_claimed=false is sticky. Package measured_j appears only when a labeled meter returns a reading.

3. Companion laws and OpenIE map

Four studies on this catalog compose. They do not collapse into one paper.

Role	Study	Owns
Navigation	Mixture of Limits	Which gear closes: Lookup → Formula → Solver → Model LAST. Floors: Vol, grammar, energy estimate, certificate, settle-refuse.
Commit	Notational Intelligence as Commit Law	propose → certify → commitrefuse → receipt.
Economic Reality of Satiation	Satiation and Scarcity after Free AI	Stop when Vol is zero on completeness (C(z)), or when budget / policy refuse fires.
Energy budget / metabolic embodiment	This paper (Metabolic Intelligence)	Budget envelope and actuators: tag sleep/heater; campus OpenADR/ MQTT levers; (J(mlq)) obtain-router.

```

flowchart TB
  NAV["Navigation<br/>Mixture of Limits"]
  COM["Commit<br/>Notational Intelligence"]
  ECO["Economic Reality of Satiation"]
  MET["Energy budget<br/>Metabolic Intelligence"]
  NAV --> COM --> ECO
  MET -. -> |binds floors at tag and campus| NAV
  MET -. -> COM
  MET -. -> ECO

```

3.1 Relation to Mixture of Limits

Mixture of Limits owns *which gear closes*. Metabolic Intelligence supplies the **μ on hardware H** those gears spend. The cascade Lookup → Formula → Solver → Model LAST is the impedance story; enzymes are a budget-native recognition leaf near Formula / cheap Solver; campus Flash routing and speed dials are empirical satiation of tok/s and model size. Soft-ref path: board_synth_claimed=false; estimates ≠ measured_j.

3.2 Relation to Notational Intelligence

Notational Intelligence owns the irreversible commit shape. Metabolic Intelligence does not certify proofs. When a campus lever sheds load or a tag refuses a heater pulse, the *permission* to act remains a Notational / WCA concern; this study owns the energy predicate those commits spend. Logical Intelligence Lean-certified commit is the strongest *peer* certify mechanism (Aleph / PutnamBench materials); it is constraint energy, not grid joules.

3.3 Relation to Satiation

Satiation owns Economic Reality of Satiation: stop when completeness (C(z)) holds and Vol is zero, or when budget refuse fires. Metabolic Intelligence makes that stop **executable**:

- Tag: skip the next heater pulse when context surprise is below threshold and lifetime budget is binding.
- Campus: refuse ascending to a larger model or higher speed when (J) is minimized for the stake.
- Civic: free-at-margin digital inference must not be misread as a licence to burn grid megawatts past done.

3.4 OpenIE map

Surface	URL	Relation
Research	research.openie.dev	This study is /papers/mei/. Companions: Mixture of Limits, Notational Intelligence, Satiation.
Stack	stack.openie.dev	

Surface	URL	Relation
		Family map; MEI is embodiment, not another stack card.
Compute	compute.openie.dev	Primitive table the cascade navigates; metabolic actuators spend primitives under μ on hardware H.
Synthesis	synthesis.openie.dev	Cost surface ($E(x)=(p)(p,H)$); this study extends obtain cost as ($J(mlq)$).
Verify	proof.openie.dev	Cousin energy law for verified artifacts.
Product	klere.ai	Klere — product home of the MEI class. This study owns the law/class; Klere owns embodiment. Public site is thin access framing (not a feature list). Tags / campus / stack named in §10 are Klere surfaces or roadmap.

Living figures: </living/mei/> (Manim clips + pack charts + teaching tables). Klere positioning: </about/mei/>. Soft-ref path keeps package energy Estimated unless Metered.

4. Standing obtain rule and the ($J(mlq)$) synthesis router

Standing rule (David Charlot): **Don't calculate what can be looked up. Don't guess what can be calculated. Don't spend hard-guessing effort on an easy guess.**

The OpenIE synthesis surface prices computing an answer:

$$[E(x) = _ \{p P(c(x))\} (p)(p,H)]$$

Metabolic Intelligence raises resolution on *how each input is obtained*. For need (q) feeding decision (y), mechanism (m) has cost:

$$[J(mlq) = E_m + S_q^2 _m(t)]$$

$$[_m(t) = (_m^2 + _q t)(1 + 3p_m), m^* = _m J(mlq)]$$

Symbol	Meaning	Evidence
(E_m)	Energy of the obtain mechanism	Stack formula on the mechanism's primitives
(S_q)	Elasticity (dy / dq)	

Symbol	Meaning	Evidence
		Calculated by central differences; never assumed
()	Joules at stake in the decision	Decision scale (one request; campus-year)
($_m$, $_q$, t , p_m)	Uncertainty, staleness rate, age, zone-misjudge chance	Looked-up spread, derivation, or labeled guess

What falls out as law of the function, not as slogan:

- Lookups dominate when they measure the same quantity.
- Cheap crude options win when (S_q^2) is small.
- Required precision: $(^* =)$.
- Refresh a cache at $(t^* = 2E / (S_q^2))$.
- The estimator of $(J(mlq))$ is Lookup or Formula. Its cost is capped so meta-compute cannot eat (B). If the estimate exceeds the cap, take the default gear. Do not run a heavy estimator.

Ledger outputs (synthesis pack; math + sourced ledger; **not** board measured $_j$):

Output	Point	p10–p90	Measure next
Wh per V4 Pro answer at today's speed (GB300)	0.20 Wh	0.044–0.68	Meter one GPU pool for a day
Work-per-MW gain, speed matching human chat	1.92×	1.19–2.93	Pro speed; human share of tokens
Same gain on Vera Rubin	2.0×	1.2–3.4	Metered Rubin chat curve

Fused public proxies for output length (400 / 630 / 1,115 tokens) give 655 at log- σ 0.585, no tighter than a 500-token guess. The router keeps the guess for campus planning and takes the lookup per request. Only the gateway's own counters settle it. Evidence class: **model + looked-up + calculation**.

```

flowchart LR
  L[Look up] --> C[Calculate]
  C --> M[Model]
  M --> G[Guess]
  J["argmin J(m|q)"] --> L
  J --> C
  J --> M
  J --> G

```

5. Digital enzymes

A **digital enzyme** is a silicon recognition unit with a binding site, a product, an allosteric context switch, Hill cooperativity, an abundance that acts as the learned weight, and a cost for every binding check. A bit-enzyme variant uses no multiplies. The unit is algorithmic compute under a local energy cost, not a smaller transformer.

Result	Number	Evidence class	Where
Instructions / decision, Cortex-M4, 12 features	Enzymes + early exit 1,723; int8 MLP 2,451; first enzyme pool 6,948	Emulator-measured	mcu/
Accuracy, same test (3 seeds)	Enzymes 82.9 / 86.9 / 88.7%; MLP 82.2 / 78.0 / 87.4%	Real protocol on simulated features	benchmarks/
Late-life accuracy, physics-invariant features (days 90–180)	LDA 94.9%, MLP 93.4%, enzymes 90.0% (from 80.8%), Tsetlin 89.1%	Simulation + classifier	benchmarks/
SmellNet, 50 foods, 5 sequential groups, no revisit	Enzymes 77.7%; MLP 18.4%	Real dataset protocol	benchmarks/
SmellNet cross-day	MLP 85.5%; enzymes 79.8% (same on Cortex-M4)	Real dataset + MCU	benchmarks/

Class properties, grounded:

- **Physics-informed.** Drift-invariant / physics-informed features added about 9–11 points of late-life accuracy for enzymes, bit enzymes, and Tsetlin machines. Sensor kinetics and humidity/heater physics are first-class inputs, not post-hoc regularizers.
- **Sequential / super learning.** Enzymes keep usable accuracy while learning foods one group at a time without revisiting. The MLP collapses under the same protocol (77.7% vs 18.4%). That is the super-learning property under a budget: absorb new context without a full pretrain replay.
- **Test-time inference.** Binding checks, early exit through the family index, and allosteric context switches are serve-time decisions under abundance and cost.
- **Not pre-training required** as default. Abundance updates and family indices learn locally. Large generative pretrain is not the substrate of the tag classifier.
- **Zero-shot under grammar.** When a Lookup or Formula leaf covers the context (known context state, known refuse), the cascade closes without opening a generative Model leaf. Mixture of Limits states that order; enzymes supply a budget-native leaf when Model is residual. Open entropy goes to Model LAST. That order is the architecture.
- **Capacity.** Family index, early exit, and abundance decay are the capacity control. The sequential learning bound is that control. Garbage collection is abundance decay.

Chicken-spoilage evaluation (Anwar & Anwar 2025) has **no licence**. Results from that dataset are **internal only, not published**. Soft-ref path: board_synth_claimed=false.

6. Cold-chain tag budget

On a gas tag the energy goes into **heating and sleep**, not into classifying. One gas reading costs about 0.5–27 mJ depending on the part (datasheet). A classifier decision costs about 92 nJ on an STM32U3 (CoreMark 15.5 $\mu\text{A}/\text{MHz}$ at 3.3 V; **calculated** from datasheet, not board package measured_j).

Metabolic schedule (about 9.6 readings/day), October 2026 parts:

Term	Power	Evidence
Heater (BME690 metabolic)	$\sim 3.0\ \mu\text{W}$	Datasheet + schedule model
MCU sleep (STM32U3)	$\sim 6.9\ \mu\text{W}$	Datasheet
Tag total	$\sim 10\ \mu\text{W}$	Sum of schedule model
CR2032 life	$\sim 8.1\ \text{years}$	Capacity model
Same tag on nRF54L15 sleep	$\sim 5.1\ \mu\text{W}$	Datasheet swap

Correction (honesty): earlier prose said “the heater is the cost.” True for frequent readings. On the metabolic schedule the **MCU sleep floor** is the larger term. Choose the MCU for sleep current. Innatera Pulsar-class neuromorphic parts publish audio/radar milliwatts, not gas-tag microwatt schedules; they are peers on the post–von Neumann map, not drop-in tag winners today.

Drift upload is a scheduled act inside that same sleep and heater discipline, inside the CR2032 envelope. It is not an extra drain beside the envelope. This study does not publish a separate radio joule. Heater duty is thermal feedback inside the same metabolic control. The published $\sim 10\ \mu\text{W}$ class figure stays datasheet plus schedule model.

Market timing (rules, not hype): FDA proposed FSMA 204 compliance move to 20 Jul 2028; Congress barred earlier enforcement. EU PPWR applies since 12 Aug 2026. These dates are the **market clock** for cold-chain observability buyers, not a product launch slogan.

7. Campus energy per answer

Method (curves pack): energy per answer $\approx$ output tokens $\times$ all-in kW per GPU $\div$ output tokens/s per GPU.

- **Throughput:** NVIDIA AI Configurator 0.12.0 (measured GPU kernels; SGLang; NVFP4). Evidence: **measured-kernel model**.

- **Power per GPU from InferenceX** (tokens/s per chip ÷ tokens/s per MW): GB300 2.12 kW, B300 1.90, B200 1.71, Vera Rubin 3.30, MI355X 2.09. Evidence: **calculated from published measurements**.
- **Served speeds (Artificial Analysis medians)**: V4 Pro 79 tok/s/user, V4 Flash 125, Qwen3-VL 48.5. Evidence: **looked up**.

Workload (GB300 unless noted)	Energy	Evidence class
Chat, DeepSeek V4 Pro (2k in / 500 out)	0.32 Wh at 50 tok/s/user; 0.92 at 79; 4.57 at 159	Curves method
Synthesis ledger best estimate today	0.20 Wh (p10–p90 0.04–0.68)	Ledger fuse
Chat, V4 Flash	49 mWh at 50; 180 at 159	Curves
Agentic turn, V4 Flash (32k in, 28k cached / 1k out)	0.16 Wh at 96	Curves
Video clip, 8 s 720p (Wan 2.2)	390 Wh	Curves / model

Plateau levers (field + model, not board measured_j):

- Vera Rubin NVL72 moves the knee of the speed-energy curve from about 100 to about 175 tok/s/user (InferenceX / MLPerf; **measured** field reports). Speed matching human chat is worth about 2.0× on Rubin vs about 2.9× on GB300 (ledger).
- Flash-first routing (V4.1 Flash answering most human chat) plus speed matching: about 2.3–4.4× more work per MW (efficiency pack; **model + looked-up**).
- Deeper speculative decoding (accept length 2.8 modeled): about 1.55× cut in today’s base-mix energy per output token; speed matching then adds only about 1.4–1.8×.

These are **test-time** levers under a live budget: which model answers, how fast it streams, how much draft speculation to accept. They are Metabolic Intelligence at campus scale, not a new pretrain run.

Bands matter. Point estimates without p10–p90 invite false precision. Soft-ref path: `board_synth_claimed=false`.

8. Metabolic grid service (resource-optimized central)

The service makes a **budget-capped** 100 MW AI plant — a **resource-optimized central datacenter**, not an uncapped hyperscale campus — follow grid caps. Lever order on deferrable work: **pause deferrable work** → **slow per-user speed** → **route to a smaller model** → **lower reasoning effort** → **battery** → **shed**. Those levers do not degrade the certified answer. Certified and interactive work runs to close or to refuse. Deferrable work is what pauses. A breakthrough task is not aborted by a linear spend line. A contract declares the invariants (`never_lower_precision`, `max_deferral`). Grid follow stays inside the contract. Shed is last and only outside the invariant. The answer that commits is the best one. Route and effort move deferrable work. They are not a license

to commit a worse answer. The withdrawn claim that Flash routing costs no quality stays withdrawn. The power-model bias loop is live thermal control on this service (plant model: learned +5 MW on a hot day). Grid-following lean operation is how the class stays inside shared power and water ecosystems without requiring a private dedicated cooling/power campus. That framing is a design consequence of the envelope; it is not a measured water-liter claim. Soft-ref path: board_synth_claimed=false.

Architecture (datacenter/service/):

- **Grid in:** OpenADR 3.1 VEN (OAuth2, polling, DEMAND reports), tested against real openleadr-rs VTN. Evidence: **real protocol**.
- **Backstop:** DSX Flex ISV over MQTT, CloudEvents, JWS ES256. Evidence: **real broker + schemas**.
- **Actuators:** Dynamo planner GPU budgets; metabolic gateway (in-flight caps, routing, effort, x-consumer speed tiers); Kueue / SGLang pause; Redfish node sleep; battery PCS.
- **Learning:** per-pool speed curve by recursive least squares; power-model bias loop (learned +5 MW on a hot day in the plant model).
- **Plant:** campus model. Evidence: **model**. Protocols and schemas: **real**.

Scenario (100 MW GB300 campus)	Time over cap	Interactive served	Note
A–C: 60/70% day; notice / feedforward / no notice	0	100%	Notice mostly saves battery
D: 90/95%, fast campus (159 tok/s/user)	0	100%	All knowledge chat routed; effort low
E: 90/95% without node sleep	96 min	1%	Idle GPUs alone exceed a 10 MW cap
L–M: today's looked-up speeds, 60/70 and 90/95	0	100% / 52.6%	Speed already spent cannot be spent again
N: no notice, no battery (PJM-style 10 min)	7.5 min	99.8%	Under cap within 225 s
O: 60% cut for 10 h	0	100%	Deadlines kept
P: new cap every 5 min for 3 h	0	98.8%	ERCOT-style
Q: recovery ramp 2 MW/min	0	100%	Max rise 2.9 MW/min vs 29–47 unlimited
R: 90% cut for 10 h	0	13%	Video deadline vs interactive (policy)

In every run: no actuator call failed; VTN accepted all OpenADR reports; no DSX message rejected. Headline: **zero time over the grid cap whenever node sleep is available**. Correction honesty: “100% served through a 90/95% cut” assumed a fast campus and one curve; with per-pool curves it is about 94%, from today’s speeds about 53%. The **cap held** in all of them.

AEMA (Emerald AI, Google, NVIDIA; 16 Sep 2026) is a field alliance; **no published AEMA protocol spec** as of the 1 Oct 2026 radar. Dynamo 1.5 power-aware scaling (18 Sep 2026) is a deployment annotation, not obtain-routing. OpenIE wires those actuators *inside* Mixture of Limits intelligence.

9. Economic argument: joules as scarce resource

9.1 Free at the margin for digital chores

[Satiation](#) states the economic fact: published money price of a fixed inference performance has fallen sharply (Epoch AI series cited there). For digital chores whose output is information and whose completeness test is finite, synthesis can be **free at the margin** in money terms. That phrase does **not** mean joules are free, that a plant may move without a certificate, or that care labor has been automated away.

Metabolic Intelligence accepts that money-price collapse and asks the physical question Satiation leaves to the envelope paper: *who spends which joules after the chore is done, and who is refused?*

9.2 Two scarcities — and the ecosystem that uncapped campuses pull

Scarcity	What binds	Who feels it
Money price of tokens	Collapsing for fixed benchmarks	Buyers of API seats; can look “free”
Grid / actuation joules	Import caps, interconnection queues, heater mJ, battery cycles	Ratepayers, farms, cold-chain operators, campus neighbors
Ecosystem support (water, dedicated power, hyperscale cooling)	Uncapped campus design that presupposes private cooling loops and generation	Shared basins, ratepayers, neighbors of hyperscale builds

People should not have to fight over energy to eat or to use AI. The civic framing is not marketing copy. It is the reason tags that watch food must last years on a coin cell, and lean centrals that serve answers must shed watts when the grid asks—without lying about meters.

Design consequence (not a liters-saved claim). Metabolic Intelligence targets energy budgets so low that dedicated water plants, dedicated power plants, and hyperscale cooling loops are **not required**. Evidence in this pack is the envelope itself: ~10 μ W tag schedules (datasheet + schedule model) and a grid-following lean campus under OpenADR/DSX (real protocols; plant model). Do **not** invent gallons or liters of water saved, nor upgrade the claim to board measured_j. The statement is class design + pack evidence, not a field water meter. Soft-ref path: board_synth_claimed=false.

9.3 Work-per-MW and Wh/answer as estimates with bands

Quantity	Point	Band	Evidence	Do not claim
Wh / V4 Pro answer (today, GB300)	0.20 Wh	0.04–0.68	Ledger fuse	Board measured_j
Work-per-MW vs today (speed match)	1.92×	1.19–2.93	Efficiency + looked-up speeds	Universal 4–12×
Same on Vera Rubin	2.0×	1.2–3.4	Ledger transfer	“30×” without speed condition
Flash-first + speed match	2.3–4.4×	model band	Efficiency P4	“Routing costs no quality”

These are **estimates with bands**. Soft-ref path: board_synth_claimed=false. Metering one GPU pool for a day is the highest-Vol next measurement in the synthesis ledger.

9.4 Satiation stop as economic actuator

When completeness ($C(z) = 1$) and Vol is zero, further Model spend is a cost with no increment on the written objective. Metabolic Intelligence implements that stop as:

1. **Obtain refuse:** do not open Model when Lookup/Formula closes (J).
2. **Speed refuse:** do not stream above human-read rate when the consumer is human (x–consumer tiers).
3. **Grid refuse:** pause / shed when import cap binds, ordered by least service lost per MW.
4. **Tag refuse:** skip heater when surprise is below threshold and lifetime budget binds.

Stop is success. That is Satiation’s Economic Reality; this study shows the actuators.

9.5 Landauer as labeled estimate, not a cheat

Landauer’s ($k_B T$) appears on analytical receipts as a **labeled estimate**. It is not a claim that OpenIE boards operate at the ideal floor. Horowitz (2014) places practical CMOS far above that floor. The economic argument needs only the weaker statement: joules remain scarce after tokens become cheap; schedules and meters are how that scarcity is governed.

9.6 Rebound, care, and the civic bound

Direct rebound (Gillingham, Rapson, and Wagner, as treated in the Satiation companion) is the extra use of a service when its effective price falls. Free-at-margin tokens invite rebound in engagement metrics. Metabolic Intelligence does not estimate a structural rebound parameter. It installs **hard stops**: completeness refuse, budget refuse, and grid cap. Those stops are design rules, not fitted elasticities.

Care and food are not software goods. Baumol-style cost disease (cited in Satiation) warns that sectors with weak productivity growth absorb expenditure even when token prices fall. Cold-chain tags exist so food loss and food energy are measured under a coin-cell envelope—not so a generative model can narrate spoilage without a sensor. Lean centrals that shed watts on OpenADR signals exist so ratepayers are not forced into a false choice between “AI” and “keep the lights on,” and so shared water and power ecosystems are not presupposed as private hyperscale plants. The civic bound is: **eat and compute under shared joules, with meters and schedules, without invented board claims or invented liters saved.**

9.7 Cost surface from synthesis to metabolic spend

OpenIE synthesis prices an answer as $(E(x) = (p)(p,H))$. Metabolic Intelligence adds:

Layer	Object	Scarcity coordinate
Synthesis	Primitive mix on hardware H	Analytical / looked-up μ
Obtain router	$(J(mlq))$ over Lookup→Guess	λ at stake + variance
Tag schedule	Heater + sleep + radio	μ W continuous; CR2032 life
Lean campus service	Ordered levers under $(P_{\{\}})$	MW import (capped); interactive served

The same standing rule threads all four layers. Soft-ref path keeps every layer’s joules labeled until Metered.

9.8 Resource-optimized deployment vs uncapped hyperscale

Deployment	Energy budget posture	Ecosystem support presupposed	MEI stance
Edge (tag)	μ W continuous; coin-cell life	None beyond the cell and board	In scope — proven pack for schedule / enzyme evidence
Resource-optimized central	MW import cap ; grid-following levers	Shared grid; no private uncapped cooling/power campus required by class design	In scope — proven pack for OpenADR/DSX + scenarios
Uncapped hyperscale campus	Unlimited import until interconnection / cooling bind	Dedicated cooling, dedicated generation, large water uptake	Out of class — not the MEI design target

The market value of the class is not “we save N liters.” It is: buyers who need answers under an explicit envelope get a deployment that **does not require** the hyperscale support plant. That is settled voice for the economic argument: design consequence + pack evidence; estimates $\neq$ measured_j.

10. Market value proposition

10.1 Who buys what

Vehicle. Buyers purchase through **Klere** (klere.ai) — the product home designed around Metabolic Intelligence. This paper sells the class and the evidence packs; Klere is the commercial vehicle. Do not invent shipped SKUs from the thin public site. Sell **edge** and **resource-optimized central** — not an uncapped hyperscale campus build.

Buyer	What they buy (via Klere)	Why now (market clock)	Evidence class for timing
Cold-chain / food ops (edge)	Microwatt gas tags + enzyme recognition under metabolic schedule	FSMA 204 enforcement barred before 20 Jul 2028 (FDA proposed); EU PPWR since 12 Aug 2026	R (rules)
Resource-optimized central / colo / lean campus energy ops	Grid-following lean service: OpenADR 3.1 VEN + DSX Flex MQTT + ordered levers — not uncapped hyperscale cooling/power plants	Emerald DSX Flex commercial path (1 Jun 2026); AEMA alliance (16 Sep 2026, no spec yet)	C/V
Software / platform buyers	Enzymes + (J(mlq)) router + metabolic gateway (speed tiers, route, effort)	Flash-first routing and Dynamo power annotations make test-time levers native	M/V

10.2 Product slices (honest scope)

Label each slice as **proven pack** (evidence in this study) or **Klere roadmap** (product packaging not claimed shipped). Klere is the vehicle; this paper is not a SKU sheet.

Slice	What it is	Proven pack (this study)	Klere roadmap (not claimed shipped)	Ecosystem posture
Edge / cold-chain tags	Enzyme recognition + metabolic heater schedule on coin-cell MCUs	MCU instruction table; SmellNet sequential protocol; tag budget script (~10 μ W class, datasheet + schedule model)	Commercial tag SKU / field install / device ops	μ W envelope → dedicated cooling/ water plants not in design
Resource-optimized central	Grid-following lean campus under import cap	OpenADR 3.1 VTN + DSX Flex MQTT demos; scenario	Commercial flexibility / operator product	Shared grid; no private uncapped cooling/power

Slice	What it is	Proven pack (this study)	Klere roadmap (not claimed shipped)	Ecosystem posture
		table A–R; ordered levers		campus required by class design
Obtain / stack surface	(J(mlq)) router + metabolic gateway under Mixture of Limits	Router math; synthesis ledger bands; soft-ref companions; standing obtain rule	Packaged gateway / DX beyond mol.yaml / mol run	Software spend under explicit envelope; Model LAST

1. **Edge pack.** Digital enzyme / bit-enzyme classifiers; family index; physics-informed features; tag budget model for BME690 / ZMOD4410 / SGP41 + STM32U3 / nRF54L15. Not a neuromorphic gas-odour nJ claim (shipping figures **not found**).
2. **Resource-optimized central / lean campus service.** OpenADR 3.1 + DSX Flex–shaped control; gateway with x–consumer speed tiers; plant-model tested scenarios. Plant is a model; protocols are real. Sold as grid-following lean central — **not** as an uncapped hyperscale campus that requires dedicated water/power plants.
3. **Obtain stack.** (J(mlq)) router as Mixture of Limits embodiment; satiation stop wired to refuse codes.

GTM slices (who first). Sequence is engineering, not hype: (A) publish the class with soft-ref honesty → (B) show protocol demos to lean-campus energy buyers → (C) meter one GPU pool before any Wh/answer upgrade → (D) licence edge pack into Klere packaging when ops buyers are ready → (E) wire Dynamo / DPS when vendor hooks exist. Do not sell meters you do not have; do not invent liters saved.

10.3 Competitive differentiation

Vs transformer hyperscale (primary market foil). The default build is still: pretrain a frontier transformer, then serve tokens under interconnection and cooling that assume an uncapped campus. That path pulls dedicated generation, dedicated cooling loops, and large ecosystem water/power uptake. Klere / MEI refuses that as the design target. The product thesis is **edge + resource-optimized central**: the envelope is the condition of the superior answer; Model LAST; levers that refuse or defer rather than degrade the certified answer; coin-cell tags. Competitive claim that is allowed: *hyperscale support plants are not required by design at these envelopes*. Competitive claim that is forbidden without meters: *we save N liters / N MW of cooling water, or Wh/answer as board measured_j*.

Peer	Their axis	Metabolic Intelligence / Klere difference
Transformer-only / uncapped hyperscale stacks	Pretrain → serve tokens under unlimited import + dedicated cooling/power	Envelope-first; Model LAST; edge + resource-optimized central;

Peer	Their axis	Metabolic Intelligence / Klere difference
		hyperscale support plants not required by design
Logical Intelligence	Constraint energy + formal commit	Joule envelope + obtain routing + dual scale; LI energy $\neq$ grid joules
World Labs / V-JEPA	Spatial / latent world models	Budget-native enzymes + campus grid service; latent optional; no-pretrain scoped to enzyme path
Pure neuromorphic vendors (e.g. Innatera Pulsar)	Efficient non-transformer silicon at edge	Cousins at edge; lack campus (J(mlq)) + OpenADR thesis; Pulsar mW audio/radar $\neq$ tag μ W schedule
Dynamo / DSX / AEMA	Power and grid actuators	Actuators inside Mixture of Limits intelligence; AEMA has no published spec yet
Tsetlin / HDC	Efficient propositional / hypervector silicon	Edge cousins; measured Tsetlin ASIC 8.6 nJ/MNIST frame (Tunheim et al., Jan 2025); lack campus thesis

Honest roadmap vs proven pack. If a buyer asks “what works today?”, answer with the pack rows (emulator instructions; OpenADR/MQTT demos; schedule model; ledger bands). If they ask “what does Klere ship?”, answer roadmap and do not upgrade evidence class. Soft-ref path: board_synth_claimed=false.

10.4 What not to sell without meters

Do not sell Wh/answer, μ W lifetime, or $\times$ work/MW as board-measured without a named meter. Do not sell invented **liters / gallons of water saved** — state ecosystem-resource avoidance as a **design consequence** of the budget-native class + pack evidence (μ W tags; grid-following lean campus). Do not sell “routing costs no quality” (Flash leads AA composite; trails ~ 1.5 GPQA). Do not sell AEMA latency numbers. Do not publish chicken-spoilage Anwar accuracy (no licence). Soft-ref path: board_synth_claimed=false.

10.5 Buyer journeys (three packs)

Vehicle for all three journeys: **Klere** (klere.ai). Research study = class + evidence; Klere = product embodiment. Proven pack vs roadmap is labeled per journey.

Journey A — Cold-chain operator (Klere edge / tags surface). Problem: FSMA 204 / PPWR timing raises the cost of blind legs in the chain; coin-cell tags must last years. Offer via Klere: metabolic schedule (~ 10 μ W class on Oct 2026 parts, datasheet + model) + enzyme recognition with

sequential learning for new SKUs without full MLP retrain. **Proven pack:** MCU instruction table; SmellNet sequential protocol; tag budget script. **Klere roadmap:** commercial tag SKU / field install packaging. Not offered: Anwar chicken accuracy; neuromorphic gas nJ; board measured_j.

Journey B — Resource-optimized central / lean campus energy lead (Klere campus surface).

Problem: interconnection and peak stress; AEMA / DSX language is rising; operators need levers that preserve interactive work **without** building an uncapped hyperscale cooling/power campus. Offer via Klere: OpenADR 3.1 VEN + DSX Flex MQTT backstop + ordered levers with scenario pack (zero time over cap with node sleep). Ecosystem-resource avoidance (water / dedicated power / hyperscale cooling) is a design consequence of the envelope — not a liters-saved pitch. **Proven pack:** real VTN / broker tests; scenario table A–R; recovery-ramp redesign note. **Klere roadmap:** commercial flexibility product and operator SKU. Not offered: AEMA latency/ramp numbers; plant watts as meter readings; invented water liters; “100% served through 90% cut” as universal.

Journey C — Platform / stack buyer (Klere stack surface). Problem: serving stacks route models but do not price obtain zones; transformers stay default. Offer via Klere: (J(mlq)) router + metabolic gateway (x–consumer speed tiers, route, effort) under the Mixture of Limits product path, embodied as Klere stack surfaces. **Proven pack:** synthesis ledger bands; efficiency pack work-per-MW ranges; standing obtain rule; soft-ref companions. **Klere roadmap:** packaged gateway / DX productization beyond mol.yaml / mol run. Not offered: Wh/answer as measured_j; “routing costs no quality.”

10.6 Pricing and packaging honesty

Claim in a pitch	Allowed form	Forbidden form
Tag lifetime	“~8 years on CR2032 under stated schedule (capacity model)”	“8.00 years measured on shelf”
Enzyme vs MLP	“1,723 vs 2,451 instructions (emulator)”	“30% less board joules”
Campus flexibility	“0 time over cap with node sleep in closed scenarios (plant model + real protocols)”	“Guaranteed 0 over any grid event”
Work per MW	“1.92x point, 1.19–2.93 band (ledger)”	“2x guaranteed on your fleet”
Differentiation vs LI	“joule envelope + dual scale vs constraint energy”	“we beat Kona on proofs”
Ecosystem resources	“dedicated water/power/hyperscale cooling not required by design (μ W tags; lean grid-following central)”	“saves X liters / MW of cooling water” without a meter

10.7 Go-to-market sequencing (engineering, not hype)

Sequencing is for **Klere** as product home; this paper remains the class publication.

1. **Publish the class** (this paper + SOTA brief) with soft-ref honesty; point product inquiries to klere.ai without overclaiming the thin public site.
2. **Ship protocol demos** (OpenADR/DSX test harness) as interoperability evidence (proven pack → Klere campus roadmap).
3. **Meter one pool** (highest Vol ledger item) before any Wh/answer sales claim upgrades evidence class.
4. **Licence edge pack** into Klere packaging and seek chicken-dataset permission only if publish is required—until then keep Anwar numbers internal.
5. **Wire Dynamo 1.5 / DPS** as live actuators when vendor hooks are available; until then keep them noted, not claimed live.

11. SOTA at the metabolic frontier

11.1 Class definition

Metabolic Intelligence is the budget-native class: the metabolic envelope is the physical condition under which the superior answer is obtained. It does not trade the certified answer down to save joules, and it does not take uncapped transformer FLOPs or uncapped hyperscale import as the coordinate. Architecturally it sits with Logical Intelligence—style moves *past* basic transformers (energy-based scoring and constraint commit, latent world models, non-autoregressive or non-attention primitives), but binds the envelope at **two scales at once: edge** (microwatt tags) [?] **resource-optimized central** (budget-capped, grid-following lean campus) — not the uncapped hyperscale campus that presupposes dedicated water, power, and cooling plants. The unit of computation is the **digital enzyme** (recognition, allosteric context gating, abundance as weight, local learning and decay, costed binding check) and related algorithmic compute, not ever-larger nets. Ecosystem-resource avoidance is a **design consequence** of those budgets + pack evidence; it is not a measured liters-saved claim.

Class properties (stated as operating claims; peer overlap cited honestly below):

Property	MEI grounding in this pack	Honest peer overlap
Super / sequential learning	SmellNet enzymes 77.7% vs MLP 18.4% across five groups without revisit (real protocol)	AdaTM continual Tsetlin; classical Hopfield / DenseAM pattern add. Do not claim universal end of catastrophic forgetting.
Test-time inference	Binding, early exit, (J(mlq)) route, campus lever dial under live budget	DenseAM energy descent; Hopfield recall; JEPA MPC planning; Logical Intelligence constraint solve. Test-time ≠ free or pretrain-free.

Property	MEI grounding in this pack	Honest peer overlap
Zero-shot under grammar	Lookup → Formula → Solver first (Mixture of Limits); refuse when covered	V-JEPA 2-AC zero-shot robot planning without task reward is a strong peer. OpenADR demos here are protocol-tested , not “zero-shot AGI.”
No large pretrain required	Enzyme / bit-enzyme online path on MCU	Tsetlin / Hebbian AM cousins. Do not attribute this to JEPA, World Labs, or Logical Intelligence frontier stacks; they pretrain heavily. Scope the claim to the enzyme path.
Physics-informed	E-nose kinetics, drift-invariant features (+9–11 late-life points); heater / sleep schedule; Landauer as labeled estimate	PINN lineage; analog EBM as physics substrate. Physics features ≠ thermodynamic optimality without meters.

Class, not a peer list. Section 2.6 absorbs four objects into Metabolic Intelligence: the physics envelope, the superior answer, digital enzymes, and the dual hardware map. The table below cites the rest. Klere ([klere.ai](#)) embodies the absorbed class on the edge and the resource-optimized central. Neighbors keep their axis. None of them is this class, and none of this is a geography. The combination here is edge and resource-optimized central, an OpenADR 3.1 and DSX Flex–shaped lean service, the (J(mlq)) obtain-router, digital enzymes, and Satiation stop, with hyperscale water, power, and cooling plants not required by design. Soft-ref path: board_synth_claimed=false.

11.2 Method taxonomy (must-cite; method-first, not geography)

The prose, URLs, and dates are in §2.6. This table is the index. Digital enzymes, the physics envelope, the superior answer, and the dual hardware map are not rows: they are the class. Checked 2 Oct 2026. Do not promote a cell into measured_j.

Method (cited only)	Canonical cite	Cite for	Do not absorb
Energy-based models / Hopfield / Dense Associative Memory	Hopfield 1982; Krotov and Hopfield, arXiv:1606.01164 (2016); Ramsauer et al., arXiv:2008.02217 (2020); Energy Transformer, arXiv:2302.07253 (NeurIPS 2023); Energy-Based Transformers, arXiv:2507.02092 (2025); analog Dense Associative Memory, arXiv:2512.15002 (16 Dec 2025); energy-based	Test-time energy descent. Attention recovered as an energy update.	Their energy is not campus joules and not the tag schedule. Amplifier time bounds are theirs.

Method (cited only)	Canonical cite	Cite for	Do not absorb
	dynamical models, arXiv:2604.05042 (2026)		
Logical Intelligence	Business Wire, 20 Jan 2026; Bodnia and Hanin, 21 Jan 2026, logicalintelligence.com/blog/energy-based-models-for-reasoning	Constraint energy. Non-autoregressive trace. Lean-checkable commit via Aleph.	Not obtain routing. Not dual deployment. Not “no pretrain.” Do not copy their benchmark figures.
V-JEPA / World Labs	LeCun, v0.9.2, 27 Jun 2022; Assran et al., arXiv:2506.09985, 11 Jun 2025; Marble, 12 Nov 2025; Atlas, 1 Sep 2026	Latent prediction and spatial world models	Large video pretrain stays on their path. Optional Model-last leaf only.
State-space corridor (Mamba, Mamba-2)	arXiv:2312.00752 (2023); arXiv:2405.21060 (2024)	Linear-time selective state. Sequence efficiency versus quadratic attention.	Not an obtain zone. Not a grid cap. Not the class.
Neuromorphic / analog / photonic	Hala Point, Intel, 17 Apr 2024; Chen et al., Nature Communications, 30 Sep 2025, doi:10.1038/s41467-025-63794-4; Ahmed et al., Nature, 9 Apr 2025, doi:10.1038/s41586-025-08854-x	Substrates on the dual hardware map (Appendix B)	Not the software class. Vendor watts are not measured_j.
Tsetlin machines / hyperdimensional computing	Granmo, arXiv:1804.01508 (2018); Kanerva, 2009; Tunheim et al., arXiv:2501.19347, 31 Jan 2025 (8.6 nJ per MNIST frame, 65 nm)	Sparse logic and associative edge cousins	Not the enzyme. Not OpenADR.
Trillium Labs	trilliumlabs.org ; Lambert note and WIRED, both 2 Oct 2026	Announced program: open post-training recipes on the frontier stack	No recipe, dataset, or release is in those sources. Not Klere.

Power-aware serving and grid flexibility stay in §11.4. They are actuators.

11.3 Positioning vs named peers

Peer	Their axis	Metabolic Intelligence difference
Logical Intelligence	Constraint energy + formal commit	Joule envelope + obtain routing + dual scale

Peer	Their axis	Metabolic Intelligence difference
World Labs / V-JEPA	Spatial / latent world models	Budget-native enzymes + campus grid service; latent optional
Dense Associative Memory / analog energy-based models	Physics dynamics for inference	Cited substrate path. Enzymes and the envelope stay this class
Dynamo / DSX / AEMA	Power and grid actuators	Actuators inside Mixture of Limits intelligence
Tsetlin machines / hyperdimensional computing	Sparse non-transformer silicon	Edge cousins. The class adds the obtain-router and the OpenADR envelope
Trillium Labs	Announced open post-training recipes (2 Oct 2026). No recipe in the launch sources	Open corridor of the frontier stack. Not this class

```

flowchart TB
    subgraph classwave [Post-transformer class wave]
        LI[Logical Intelligence<br/>constraint EBM]
        WL[World Labs / JEPA<br/>latent worlds]
        MEI[Metabolic Intelligence<br/>budget envelope]
    end
    subgraph hw [Hardware dual-map]
        VN[GPU / MCU today]
        PVN[Neuromorphic / analog CIM<br/>photonic / adiabatic]
    end
    MEI --> VN
    MEI --> PVN
    LI -. -> PVN
    WL --> VN

```

11.4 Soft peers only: serving and grid actuators

Dynamo 1.5 power-aware scaling, NVIDIA DPS, InferenceX Vera Rubin tok/s/MW curves, Emerald DSX Flex, and the AI Energy Management Alliance are **envelope actuators and field measurements**. They do not define the intelligence class. Academic joule-aware routers (PEARL, OmniRouter, Energy-Aware LRM) price which LLM answers; they do not price which obtain zone (Lookup → Formula → Solver → Model LAST) and do not bind tag?campus.

Term hunt (2 Oct 2026): **Metabolic Intelligence** as a named peer class to Logical Intelligence / World Labs was **not found** outside OpenIE / Charlot Lab energy-first framing. That is a positioning fact, not a monopoly claim on every energy paper.

11.5 What not to claim without meters

- Wh/answer, J/token, μ W lifetime, or $\times$ work/MW untied to pack evidence **M/C** or a named external meter.
- Invented **liters / gallons of water saved** (state ecosystem-resource avoidance as design consequence of budget-native class + pack evidence only).
- “Routing costs no quality” (Flash leads AA composite; trails ~ 1.5 GPQA).
- “Heater is always the tag cost” (sleep floor dominates on metabolic schedule).
- Rubin “up to 30 $\times$ ” without speed condition (gain is speed-dependent).
- AEMA / DSX Flex latency or ramp numbers (unspecified).
- Shipping neuromorphic gas/odour nJ figures (not found).
- That Logical Intelligence “energy-based” equals grid joules (different energy).
- That MEI needs no pretrain **and** matches V-JEPA zero-shot robotics (incompatible scopes; split by path).
- Chicken-spoilage accuracy in public prose (**no licence**).

Soft-ref path: board_synth_claimed=false; estimates $\neq$ measured_j.

12. Review answered by construction

Metabolic Intelligence does not bargain the answer down. Klere is the product that builds the class. The envelope is the physical condition under which the superior answer is obtained. The points below are mechanisms of that class, stated as law. Pack numbers stay in their evidence class: emulator counts, datasheet schedules, protocol tests, plant models, calculations.

board_synth_claimed=false because meters are the measurement instrument of the solution. Package measured_j is written when a meter returns. No shunt joules are invented. Chicken-spoilage Anwar numbers are not published.

12.1 A priori difficulty

The obtain router prices cost before spend. (J(mlq)) is that estimate. The estimator is a Lookup or Formula gear. Its compute is capped so meta-compute cannot eat (B). If the estimate of the estimator exceeds the cap, the router takes the default gear and does not run a heavy estimator.

12.2 Enzyme capacity

Family index, early exit, and abundance decay are the capacity control. The sequential learning bound is part of the enzyme. Garbage collection is decay of abundance. Cortex-M4 counts in §5 are emulator-measured instructions for the family-index path. They are the compute term of that control.

12.3 Non-monotonic utility

Levers do not assume a smooth utility of energy. Certified and interactive work runs to close or to refuse. Deferrable work is what pauses. A breakthrough task is not aborted because a linear spend line was crossed. The answer that commits is the best one. The envelope is not a Faustian discount.

12.4 Two ledgers

Compile-time synthesis energy and the serve-time envelope are both counted. Tag microwatts (§6, datasheet plus schedule model) are the serve-time envelope. Offline synthesis sits on the compile ledger ($E(x)$). Dual accounting is the law. Compile joules stay on the compile ledger. Serve joules stay on the serve ledger.

12.5 Contract-bounded levers

A contract declares invariants: `never_lower_precision`, `max_deferral`. Grid follow happens inside the contract. Shed is last, and only outside the invariant. The ordered levers in §8 are that law on the lean campus. They refuse or defer work that would violate the envelope. They do not degrade the certified answer.

12.6 Edge sync inside the envelope

Drift upload is a scheduled act inside the metabolic budget, under the same sleep and heater discipline as the rest of the tag. It is inside the CR2032 envelope. The published schedule (about 10 μW , datasheet plus schedule model) is the envelope those acts share. This study does not add a separate radio joule.

12.7 Energy is memory plus compute

The energy metric is ($E_{\text{}} + E_{\text{}}$). Instruction counts are a compute term only. Memory movement is in the budget. Horowitz (2014) places practical CMOS energy in data movement: on the order of 10 pJ for an internal cache access or a functional operation, and 1–2 nJ for a DRAM access, in that plenary's technology frame. Those are literature orders for the physics of moving data. They are not this pack's `measured_j`.

12.8 Thermal and process loop

The control loop is live. On the campus service, power-model bias is learned (plant model: +5 MW on a hot day). On the edge, the metabolic schedule includes thermal feedback: heater duty follows the thermal state, inside the same envelope. That loop is metabolic control.

12.9 Open entropy is Model LAST

A bounded domain is the class. Work covered by Lookup, Formula, or Solver closes there. Open entropy goes to Model LAST. That order is the architecture.

12.10 Primitive distillation

Grammars and formulas are extracted from traces and installed as Lookup, Formula, or enzyme. Human craft of those leaves is not the long-term path. Sutton's Bitter Lesson (2019) says general methods that scale with compute overtake hand-built features. This class answers that lesson by distillation into Lookup, Formula, and enzyme. It does not answer it with uncapped pretrain.

12.11 Tier meters

Energy is reported by tier. When a meter returns, the package field `measured_j` is that reading. Until then the row stays an estimate, a datasheet, an emulator count, a protocol test, or a plant model. Estimates stay estimates.

13. Corrections log

Condensed from the Metabolic Intelligence handoff corrections log. Withdrawn claims are stated plainly.

Date	Withdrawn or revised	Correction
2 Oct 2026	FSMA 204 "compliance moved to 20 Jul 2028" as settled	FDA proposed ; Congress barred enforcement before that date
2 Oct 2026	PJM IRAS "full reduction within 10 min"; effective date 12 Oct 2026	Transmission owner must start and complete reduction within 10 min of PJM quantity; effective date unverifiable, removed
2 Oct 2026	STM32U3 at 12 μ A/MHz as CoreMark	That figure is a while(1) loop at 48 MHz; CoreMark is 15.5 μ A/MHz (51 pJ/cycle; 92 nJ/decision calculated)
1 Oct 2026	"V4.1 Flash routing costs no quality"	Leads V4 Pro on AA composite (39 vs 36); trails ~1.5 GPQA points on knowledge
1 Oct 2026	Recovery ramp as fixed rising line	Lagging actuators jumped 13–31 MW/min; redesigned plan+one-step with battery on meter (2.9 MW/min)
1 Oct 2026	100% served through 90/95% cut as universal	Fast campus / one curve; per-pool curves ~94%; today's speeds ~53%; cap held in all
1 Oct 2026	Speed-matching gain 2–4 $\times$ / 4.6–12 $\times$	Baselines were guesses; looked-up speeds $\rightarrow$ 1.65–2.9 $\times$
1 Oct 2026		

Date	Withdrawn or revised	Correction
	All-in kW 2.25 / 2.0 / 2.15 (GB300/B300/B200)	InferenceX-derived: 2.12 / 1.90 / 1.71
1 Oct 2026	B300 “as good per Wh at low speed”	Withdrawn; GB300 2.9× B300 tokens/MW at 96 tok/s/user (InferenceX)
1 Oct 2026	Chat answer 5–62 mWh from InferenceX J/token	InferenceX counts total tokens (mostly cached input); replaced by AI Configurator curves
1 Oct 2026	Flash at 0.33× Pro energy/token	AI Configurator: 0.04–0.15×; routing stronger, speed dial weaker
1 Oct 2026	Edge: “heater is the cost”	Frequent readings yes; metabolic schedule → MCU sleep floor dominates
Sep 2026	Enzyme MCU cost estimated in Python	Real counts 2.3–2.9× MLP; family index → 0.70×
Sep 2026	Older systems as current SOTA; XOR as two-site MWC	Relabelled or removed

Chicken-spoilage Anwar numbers stay **unpublished** (no licence).

14. Measurement classes

Evidence classes used in this study: **real protocol**, **emulator-measured**, **datasheet**, **measured-kernel model**, **looked-up**, **calculation**, **plant model**, **estimate**. Estimates ≠ measured_j. Soft-ref path: board_synth_claimed=false.

Tier meters are how the solution reports energy (§12.11). Package measured_j is written when a meter returns. Until then the row stays in the class below. Stake order on the synthesis ledger, for the next meter readings:

1. One GB300 pool for a day (energy-per-answer band; campus-scale value of the meter).
2. Vera Rubin chat curve from a metered pool or AISimulate once Rubin-supported.
3. Gateway counters: output tokens per answer; human share (x–consumer).
4. V4.1 Flash throughput curve (the curve behind routing, in place of the V4 Flash stand-in).
5. Speculative-decoding accept length per workload from engine telemetry.

Tag figures are datasheet plus schedule plus emulator. Campus figures are curves plus plant model plus real protocols. Stage C package joules are not claimed. No invented board package joules. No invented shunt joules. Chicken-spoilage Anwar numbers stay unpublished (no licence).

15. Conclusion and relation to Mixture of Limits product

Metabolic Intelligence teaches a budget-native class: physics-informed, test-time, zero-shot under grammar, sequential / super learning without mandatory pretrain, dual-mapped to GPUs/MCUs today and to neuromorphic / analog / photonic / adiabatic substrates next. The envelope is the physical condition under which the superior answer is obtained. It is not a cheaper answer. The mechanisms are stated as law in [Review answered by construction](#). Designed for the **edge** and the **resource-optimized central datacenter** — not the uncapped hyperscale campus. At those energy budgets, dedicated water, power, and hyperscale cooling plants are **not required** (design consequence of the class + pack evidence: μW tags; grid-following lean campus — not invented liters saved). Digital enzymes prove the class at microwatts. The OpenADR / DSX Flex service proves the class at a budget-capped 100 MW lean plant. The (J(mlq)) router makes the standing obtain rule executable. The economic argument binds free-at-margin digital inference to scarce joules and shared ecosystems. The market value proposition names who buys tags, lean-campus flexibility, and the software stack **through Klere**—without selling unmetered numbers or inventing live product features from a thin landing page. The argument a skeptic must answer is [Why this is the future](#). Dismissals and their evidence class are in Appendix F. Soft-ref path: `board_synth_claimed=false; estimates  $\neq$  measured_j`.

Companions stay distinct. [Mixture of Limits](#) navigates Lookup $\rightarrow$ Formula $\rightarrow$ Solver $\rightarrow$ Model LAST and owns the product path (`mol.yaml / mol run / A1–A14`; [openIE-dev/mixture-of-limits](#)). [Notational Intelligence](#) owns certify-before-commit. [Satiation](#) owns Economic Reality of Satiation. This study owns the envelope and actuators that make those floors bind when food is watched on a coin cell and when a campus must shed watts without lying about meters.

Product surface: Klere under Mixture of Limits

Metabolic Intelligence is **designed for Klere** ([klere.ai](#)). This study does not replace the Mixture of Limits product path (`mol.yaml / mol run / A1–A14`). It supplies the envelope those floors spend; **Klere** is the product embodiment of the MEI class (edge tags / resource-optimized central / stack as Klere surfaces or roadmap — not invented shipped features; not uncapped hyperscale). The Mixture of Limits cascade decides the gear; metabolic actuators and the obtain-router decide how hard the envelope may be driven at tag sleep current and at lean-campus import meters. When Vol is zero or the budget refuse fires, Satiation and Mixture of Limits already name stop as success. This study shows that stop is executable with OpenADR reports and node sleep, that enzyme recognition can stay inside a coin-cell envelope without a transformer pretrain, and that the class does not require dedicated water/power/hyperscale-cooling plants as a design consequence of its budgets.

Soft-ref path: `board_synth_claimed=false; estimates  $\neq$  measured_j`. Living figures: [/living/mei/](#) (Manim dual-scale + obtain-router clips; pack charts; teaching tables). Investor/partner one-pager: [/about/mei/](#). PDF: [/pdfs/mei.pdf](#) when regenerated.

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Edge / TinyML / market clock

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OpenIE pack and SOTA brief

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No chicken-spoilage Anwar publish numbers. No invented citations. Numbers not tied to pack evidence M/C or a named external meter remain estimates. Soft-ref path: board_synth_claimed=false.

Appendix A. Notation quick reference

Symbol	Meaning
(B)	Budget envelope (J or W / MW)
((B))	Obtainable answers under (B)
(U(a))	Stated utility of answer (a)
(q, y)	Need and decision
(m, M)	Obtain mechanism and menu
(E _m)	Energy of mechanism (m)
(S _q)	Elasticity (dy / dq)
()	Joules at stake
(_m , _q , t, p _m)	Uncertainty, staleness rate, age, misjudge chance
(J(mlq))	Obtain cost
(C(z))	Completeness predicate (Satiation)
(E(x))	Synthesis cost surface
((p,H))	Primitive energy on hardware H
(P _{} , P _{})	Grid cap and idle import

Appendix B. Dual hardware map (operational)

Class leaf	Runs today (von Neumann)	Post–von Neumann path	Status in this study
Digital enzyme	MCU / Cortex-M4 C	Content-addressable / sparse associative	Emulator-measured instructions
(J(mlq)) router	CPU / gateway service	N/A (software)	Math + ledger
Campus levers	GPU planners, Redfish, MQTT	Analog power fabrics (speculative)	Real protocols; plant model
Latent / EBM optional leaf	GPU ViT / Lean	Analog DenseAM / memristor Hopfield	Cited peers; not OpenIE meters
Photonic / adiabatic	Lab claims	Joule-floor substrates	Cited; not MEI software class

Appendix C. Civic one-pager (for non-specialists)

1. AI answers are getting cheaper in money. Electricity, water, and food cold-chains are not free.
2. **Metabolic Intelligence** means: the energy envelope is the physical condition of the superior answer. It is not the cheapest answer. Levers refuse or defer work that would break the envelope. The answer that commits is the best one.
3. On a food tag (the **edge**), that means years on a coin cell and smart heating—not a giant model in the package.
4. On a **resource-optimized central** plant, that means following the grid when asked, keeping interactive work alive, and stopping when the chore is done — not building an uncapped hyperscale campus with its own cooling and power plants.
5. At those lower budgets, dedicated water, power, and hyperscale cooling are **not required by design**. We do not invent liters saved; we show the envelope and the pack evidence.
6. We label every number: measured, datasheet, model, or estimate. We do not invent board joules.
7. People should not have to fight over energy to eat or to use AI.
8. Product home: **Klere** ([klere.ai](#)). Research owns the class; Klere owns embodiment.

Appendix D. Glossary of definitions

Spell-outs and working definitions used in this study. First use in the body also carries a short gloss; this appendix is the lookup table.

Term	Definition
Metabolic Intelligence (MEI)	Budget-native class. The metabolic envelope is the physical condition under which the superior answer is

Term	Definition
	obtained, at the edge and the resource-optimized central. Not a cheaper answer. Not a Faustian trade.
Klere	Product home of the MEI class (klere.ai). Research owns the law/class; Klere owns embodiment. Public site is thin access framing unless a feature is labeled proven pack.
Budget envelope (B)	Explicit joule or power condition under which the superior answer is obtained. Not a quality discount.
Digital enzyme	Silicon recognition unit with binding site, product, allosteric context switch, Hill cooperativity, abundance as learned weight, and a cost for every binding check. Bit-enzyme variant uses no multiplies.
Metabolic schedule	Duty cycle for sensing and heating that meets a monitoring objective while minimizing average power (heater + sleep + radio).
Obtain mechanism	How an input is acquired: Lookup, Formula / Calculate, Solver, Model, Guess — ordered by Mixture of Limits.
Obtain-router / (J(mlq))	Rule ($m^* = {}_m J(mlq)$) with ($J(mlq) = E_m + S_q^2 {}_m(t)$).
Grid-following service	Control plane that accepts OpenADR / DSX Flex signals and applies ordered levers so campus import stays under cap.
Resource-optimized central	Budget-capped, grid-following lean campus. Not an uncapped hyperscale plant with dedicated water/ power/cooling as design defaults.
Edge / tag	Microwatt-class cold-chain or similar sensing node under coin-cell physics.
Evidence class	Label on a number: real protocol, emulator-measured, datasheet, measured-kernel model, looked-up, calculation, plant model, estimate (pack radar M/C/V/ R/D/S/P).
Proven pack	Evidence artifact present in the OpenIE metabolic handoff pack and cited with its class in this study.
Klere roadmap	Product packaging / SKU / operator surface not yet shipped as a live Klere feature; may reuse proven pack under separate evidence class.
Estimate	Number that is modeled, looked-up, or curve-derived and not a board package meter.
Soft-ref path	

Term	Definition
	board_synth_claimed=false; estimates $\neq$ package measured_j.
Value of Information (Vol)	Howard (1966): expected improvement in decision utility from an observation, net of cost.
Satiation	Companion law: stop when Vol is zero or budget refuse fires (Economic Reality of Satiation).
Mixture of Limits	Companion navigation law: Lookup $\rightarrow$ Formula $\rightarrow$ Solver $\rightarrow$ Model LAST.
Notational Intelligence	Companion commit law: certify-before-commit.
OpenADR	Open Automated Demand Response protocol (3.1 VTN used in pack demos).
DSX Flex	Emerald / NVIDIA-shaped campus flexibility MQTT service used as peer actuator shape in pack demos.
AEMA	AI Energy Management Alliance (Emerald AI, Google, NVIDIA). No published protocol spec as of 1 Oct 2026 radar.
board_synth_claimed	Repository honesty flag. False means no FPGA board synthesis/meter claim.
measured_j	Package joule field populated only when a labeled meter returns a reading (Metered).
Dual ledger	Compile-time synthesis energy and serve-time envelope, both counted. Tag μ W is serve-time. Offline synthesis is compile-time.
Contract invariant	Declared bound on levers (never_lower_precision, max_deferral). Grid follow stays inside it. Shed is outside it and last.
Primitive distillation	Extraction of grammars and formulas from traces into Lookup, Formula, or enzyme.

Appendix E. Claim ledger (proven pack vs roadmap vs estimate)

Sell these as **separate evidence classes**. Do not upgrade a row without changing its class and citation.

E.1 Proven pack (cite in sales as pack evidence)

Claim	Evidence pointer	Class
	Pack mcu/ emulator	Emulator-measured

Claim	Evidence pointer	Class
Enzyme families + early exit: 1,723 instr/decision vs int8 MLP 2,451 (matched Cortex-M4 setting)		
SmellNet sequential: enzymes 77.7% vs MLP 18.4% (50 foods, 5 groups, no revisit)	Pack benchmarks/	Real dataset protocol
Metabolic tag schedule $\sim 9.92 \mu W$ total (heater ~ 2.99 , sleep ~ 6.93) on BME690+STM32U3 model	Pack tag_budget_2026_results.json	Datasheet + schedule model
OpenADR 3.1 VTN + DSX Flex MQTT service tested end-to-end	Pack datacenter/service/	Real protocol
Ordered levers: pause $\rightarrow$ speed $\rightarrow$ route $\rightarrow$ effort $\rightarrow$ battery $\rightarrow$ shed	Pack service + study §8	Real protocol + plant model
Closed scenario set with node sleep: zero time over cap (pack headline)	Pack scenario table / charts	Plant model + real protocol
Standing obtain rule / (J(mlq)) math + ledger bands	Pack synthesis/ + study §4	Calculation + looked-up
Soft-ref companions (Mixture of Limits / NI / Satiation)	Catalog papers	Research study

E.2 Klere roadmap (not invented shipped features)

Surface	What is proven underneath	What is still roadmap
Edge / tag SKU	Enzyme + schedule pack evidence	Klere packaging, ops licence, field SKU
Lean-campus operator product	OpenADR/DSX demos + scenario table; contract invariants are class law (§12.5)	Operator console and multi-site packaging
Obtain / stack gateway	Router math, soft-ref mol companions, speed-tier synthesis	Packaged gateway DX beyond research surfaces
Dynamo 1.5 / DPS live actuators	Vendor docs as envelope peers	Live wired power-cap actuators in Klere

E.3 Estimate / labeled model (not board measured_j)

Quantity	Status
Wh/answer curve bands	Curves / measured-kernel model $\rightarrow$ Estimated until Metered

Quantity	Status
CR2032 lifetime years	Capacity model from datasheet ÷ average μW → Estimate
Campus plant watts / time-over-cap charts	Plant model (protocols real) → Model
Landauer ($k_B T$) on receipts	Labeled thermodynamic lower-bound estimate
Water / power / cooling “not required by design”	Design consequence of envelope + pack evidence — not measured liters saved
AEMA latency / ramp	Forbidden — no published protocol spec
Anwar chicken-spoilage accuracy	Internal only — no licence; not published

E.4 Soft-ref lock

board_synth_claimed=false. Package measured_j only when Metered. Investor/partner prose must keep proven pack, product packaging, and estimate on separate lines. board_synth_claimed=false names the meter instrument. It does not weaken the class.

E.5 Mechanisms of the class (design law)

These rows are the construction in §12. They are not open work. Evidence class on any number stays with the pack row that measured it.

Mechanism	Law
A priori obtain	(J(mlq)) before spend. Estimator is Lookup or Formula, capped. Over cap: default gear.
Enzyme capacity	Family index + early exit + abundance decay. Decay is garbage collection.
Non-monotonic utility	Certified and interactive work closes or refuses. Deferrable work pauses. No abort of a breakthrough by linear spend.
Dual ledger	Compile-time synthesis and serve-time envelope are both counted. Tag μW is serve-time.
Contract levers	Invariants never_lower_precision, max_deferral. Shed last, only outside the invariant.
Edge sync	Drift upload is inside the metabolic budget and the CR2032 envelope. No separate radio joule published.
(E_{}+E_{})	Instruction counts are the compute term. Memory movement is in the budget. Horowitz orders are literature physics, not measured_j.
Thermal loop	

Mechanism	Law
	Campus bias learning (plant model) and edge heater feedback are the control.
Model LAST	Open entropy goes to Model LAST. That is the architecture.
Primitive distillation	Traces yield Lookup, Formula, and enzyme. Not uncapped pretrain.
Tier meters	measured_j when a meter returns. Estimates stay estimates.

Appendix F. Objection ledger

These are the dismissals that try to put Metabolic Intelligence back inside “not the future”: a nicer serving stack, a pretrain story, a hardware fashion, a peer that already finished the move, an empty market, or a pile of estimates. Each reply is one paragraph. The evidence class is what has to be denied. None of these replies invent liters, board measured_j, or chicken-spoilage numbers. None of them speak for Logical Intelligence, V-JEPA, or World Labs beyond the axis this study actually cites.

F.1 “This is just efficient inference.”

Efficient inference prices tokens, speed, or which model answers, under a transformer that is still the substrate. Metabolic Intelligence prices the envelope first, then the obtain zone. Mixture of Limits orders Lookup, Formula, Solver, and Model last. The obtain-router (J(mlq)) is calculation on that menu, not a leaderboard of large language models. Dynamo 1.5 power caps, PEARL, OmniRouter, and Energy-Aware LRM are real envelope actuators or joule-aware routers. Section 11.4 keeps them as soft peers: they do not bind a microwatt tag to a lean campus, and they do not choose an obtain zone. Denying this reply means showing a published class that already does that stack. Term hunt on 2 Oct 2026 did not find one. That is a positioning fact, not a monopoly on every energy paper. Evidence class: **class definition** plus **peer taxonomy** (study §11), not a wattmeter.

F.2 “You still need a pretrain.”

The no-pretrain sentence is scoped to the digital-enzyme path (Proposition 7). On that path, abundance and binding learn under a local cost. SmellNet sequential learning (50 foods, five groups, no revisit) is a real dataset protocol: enzymes hold 77.7% where an int8 MLP falls to 18.4%, without a full pretrain replay of earlier groups. Cortex-M4 instruction counts are emulator-measured, not joules. The same sentence is false if it is pointed at V-JEPA 2-AC or World Labs, which require large video pretrain for their zero-shot planning results, or at Logical Intelligence’s frontier stacks. Split the path or the claim is mixed and should be rejected. Evidence class: **real dataset protocol** (enzyme path only) plus an explicit **scope limit**.

F.3 “Neuromorphic hardware will eat this.”

Neuromorphic, analog in-memory, photonic, and adiabatic machines are substrates on the hardware map (Appendix B), not a rival definition of the class. Computers are hardware. The class has to run on whatever hardware spends the joules. It runs today on MCUs and GPUs. It points at post-von Neumann parts where energy-based and latent settle map more natively. Innatera Pulsar, a shipping neuromorphic microcontroller, is about 0.4–0.6 mW on audio and radar in the pack notes, far above a duty-cycled tag near 10 μ W (datasheet plus schedule model). Shipping neuromorphic gas or odour nanojoule figures were not found. A substrate can lower ((p,H)). It does not by itself choose Lookup before Model, follow an OpenADR signal, or stop when Value of Information is zero. Evidence class: **vendor** (Pulsar class) plus **datasheet / schedule model** (tag). Not a claim that neuromorphic parts are useless.

F.4 “Logical Intelligence already did energy-based models.”

Logical Intelligence is a class neighbor, not a predecessor that already owns this envelope. Kona 1.0 pilots (Business Wire, 20 Jan 2026) move constraint energy and Lean-checkable commit into product pilots. That energy is the energy of a constraint landscape. It is not campus import joules, and it is not a coin-cell schedule. Notational Intelligence, the companion commit law on this catalog, is the Open Interface Engineering surface for certify-before-commit. Logical Intelligence is the strongest peer certify mechanism this study cites. Metabolic Intelligence does not claim their proofs, their PutnamBench numbers, or their pilots. The difference on the axis that matters here is the joule envelope, obtain routing, and dual deployment. Evidence class: **vendor** (their pilots) versus **this pack** (tag schedule, OpenADR / DSX protocol, obtain-router math). Denying the difference requires equating constraint energy with grid joules.

F.5 “World Labs or JEPA is already the post-transformer future.”

V-JEPA 2 and V-JEPA 2-AC (Assran et al., 11 Jun 2025) are a strong zero-shot planning peer: latent prediction and robot planning without a task reward, after large video pretrain. World Labs Atlas (1 Sep 2026) continues a spatial world-model corridor. Metabolic Intelligence treats a latent model as an optional Model-zone leaf under Mixture of Limits, not as the enzyme substrate. Their zero-shot result is real and should not be shaved down. It also does not price a grid cap, a heater duty cycle, or Lookup before Model. Using their result to retire a budget-native class mixes two methods. Evidence class: **measured / vendor** on their axis (cited, not re-metered here) and **path scope** on ours (Proposition 7).

F.6 “There is no market.”

The market claim in this study is a buyer and a clock, not a total addressable market. Cold-chain operators face a rules clock: FDA’s FSMA 204 move is proposed, and Congress barred enforcement before 20 Jul 2028; EU PPWR has applied since 12 Aug 2026 (evidence class **R**). Lean-campus energy leads already sit in OpenADR and DSX Flex language; the AI Energy Management Alliance has no published protocol spec as of the 1 Oct 2026 radar, so this study does not sell alliance latency. Platform buyers already route models and still do not price obtain zones. Klere is the

commercial vehicle. Proven pack is what can be shown now (emulator counts, SmellNet protocol, schedule model, OpenADR and MQTT demos, router math). Klere roadmap is packaging, not a shipped catalog invented from a thin landing page. Evidence class: **rules** for timing, **proven pack** for the offer, **roadmap** kept separate. No invented revenue.

F.7 “Estimates are not watts.”

Correct, and already the rule of this study. A curve, a ledger watt-hour, a CR2032 year count, and a Landauer term on a receipt are estimates or models until a labeled meter returns a reading. Package measured_j stays empty. board_synth_claimed=false. What is not an estimate: OpenADR 3.1 and DSX Flex MQTT interoperability (real protocol), Cortex-M4 instruction counts (emulator-measured), SmellNet sequential accuracy (real dataset protocol), and vendor electrical tables used as schedule inputs (datasheet). The future argument does not need those estimates promoted to watts. It needs the envelope to be the first coordinate, and it needs the labeled rows left in their class. A dismissal that says “therefore uncapped hyperscale is the only serious future” changes the subject. Evidence class: **evidence grammar** (§2.7) and Appendix E.

F.8 “Hyperscale water and power are inevitable.”

They are inevitable only for a design whose first coordinate is uncapped import. Metabolic Intelligence refuses that design. Edge intelligence under a metabolic schedule does not contain a cooling plant. A resource-optimized central stays inside a stated megawatt cap and follows the grid, with ordered levers (pause, then speed, then route, then effort, then battery, then shed). Lemma D: if idle import already exceeds the cap, levers that leave nodes powered cannot meet a deep cut. Node sleep is part of feasibility, not a slogan. The civic sentence follows from the envelope, not from a water meter: people should not have to fight over energy to eat or to use AI. Dedicated water, dedicated power, and hyperscale cooling are not required by design at these budgets. Say that. Do not convert it into liters saved. Evidence class: **design consequence** of Propositions 1 and 5 plus the tag schedule model. Not **M** on a field water meter.

Soft-ref path: board_synth_claimed=false. Estimates are not measured_j. Chicken-spoilage Anwar numbers stay internal (no licence).